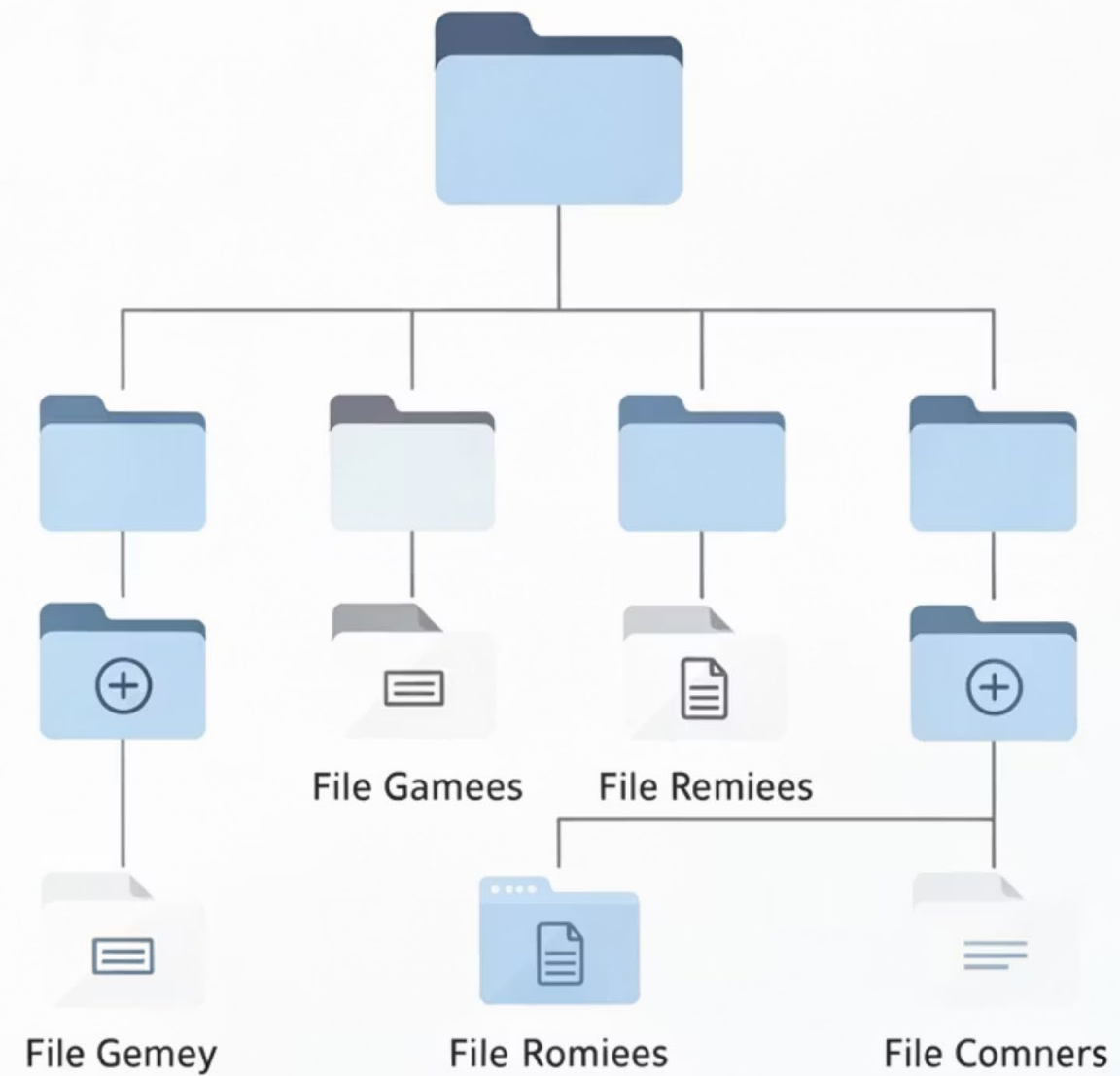




Virtual File System Simulator

SPL Team-11

Team Members

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Project Overview

This project is all about building a file system simulator, but instead of using a fancy GUI, we're going to keep it simple and work with a command-line interface (CLI).

The idea is to create a virtual file system that behaves like the one you have on your computer — managing files and directories, but in a console.

You'll be able to create, delete, move, and rename files, and navigate through directories —just like you would on your regular OS, but with a little more control and a custom shell we build ourselves.

Motivation Behind the Project

Why This Project?

Most of us use file systems every day, whether it's managing files on a computer or cloud storage like Google Drive. But we rarely get to see how it works behind the scenes.

This project lets us dive into how file systems manage files —things like directory structures, file metadata, and permissions.

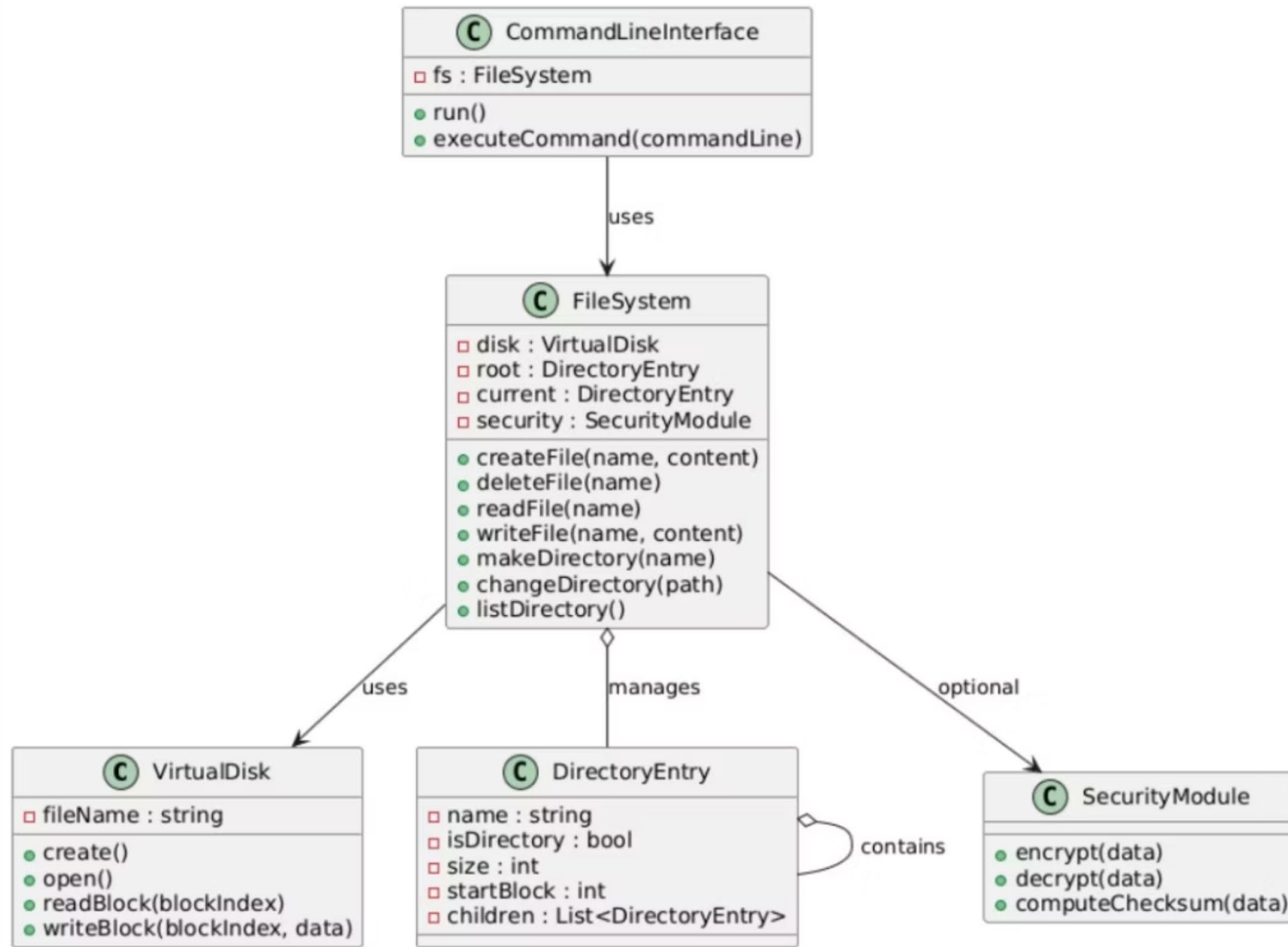
Plus, it's a great way to practise C++, object oriented programming, and using version control with Git.

Future Goals

For future version of the system, we have several features in mind:

- **Encryption with Keys** - Implement RSA or AES encryption for secure file protection with password based access.
- **Directory Tree Visualization** - Display folder hierarchy in a tree format directly in the CLI.
- **Search by Metadata** - Enable searching files by size, creation date, and modification date.

UML Diagram



Key Features of the Project

Directory Management

Create, delete, and navigate through folders

File Operations

Add, rename, move, and delete files

File encryption

Basic encryption using Caesar Cipher or XOR algorithm.

Command -Line Interface (CLI)

Control the file system using commands

Logging

Track operations like file creation or deletion

Tools and Technology



Programming Language

C++



OOP (Object Oriented Programming)

OOP principles like classes, inheritance, and encapsulation



Data Structures

Trees for directories, linked lists for file storage



Version Control

Git to manage changes as we build the project



Storage

We'll use a virtual disk file (disk.img) to store everything

Development Phases

Phase 1: Foundation

Virtual disk with block management

Phase 2: File System Structure

Superblock, inodes, and free space tracking

Phase 3: File Operations

Create, read, write, and delete files

Phase 4: Directory System

Directory management and path navigation



Phase 5: CLI

Command-line shell and commands

Phase 6: Security

File encryption and Data Integrity

Phase 7: Advanced Features

Operation logging and search

Phase 8: Testing & Polish

Testing, error handling, and documentation

Proposed 13 - Week Timeline

Week 1-4: Idea Generation and Planning

Week 7 -8: Phase 2 - File System
Structure & Phase 3 - File Operations

Week 11: Phase 6 –File encryption and
Data integrity

Week 13: Phase 8 - Testing & Polish

Week 5 -6: Phase 1 – Foundation
(Virtual Disk)

Week 9 - 10: Phase 4 - Directory System
& Phase 5 - CLI

Week 12: Phase 7 - Advanced Features

Thank You